

Ahmad Bashir

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Location: Lahore, Pakistan

PROFESSIONAL SUMMARY

Machine Learning Engineer with over 3 years of commercial experience in **computer vision**, **deep learning**, and **real-time video processing**. Specialises in **generative AI**, **diffusion models**, **3D reconstruction**, and **AI-driven video generation** using Python, PyTorch, and OpenCV. Currently working on R&D at Colossyan Inc., delivering production-grade AI avatar and video synthesis pipelines at scale. Former Research Assistant at ITU's Intelligent Machines Lab with peer-reviewed research contributions in computer vision and robotics. Proven ability to build, optimise, and deploy cutting-edge AI solutions that drive measurable business impact.

TECHNICAL SKILLS

- **Languages:** Python, C++
- **ML Frameworks:** PyTorch, TensorFlow, Diffusion Models, Stable Diffusion, NeRF, YOLO, OCR, ONNX
- **CV Libraries:** OpenCV, PCL, Scikit-Image, Pillow, TorchVision, FFmpeg
- **Data & Orchestration:** Pandas, NumPy, Matplotlib, Scikit-Learn, Apache Spark, Argo Workflows, n8n
- **Tools & Platforms:** Git, Docker, AWS (EC2, S3), Linux, CI/CD, Jupyter Notebook, Weights & Biases, TensorBoard
- **Specialist Domains:** Computer Vision, Deep Learning, Generative AI, Diffusion Models, 3D Vision, AI Avatars, Real-Time Video Processing, Neural Radiance Fields, Point Cloud Processing

WORK EXPERIENCE

- **Colossyan Inc.** Budapest, Hungary (Remote)
Machine Learning Engineer Dec 2022 – Present
 - **Avatar Pipeline Architecture & Lip-Sync:** Redesigned the core avatar generation pipeline architecture, improving **lip-sync accuracy** and overall fidelity by **20%**, directly enhancing production video realism.
 - **Large-Scale Data Processing:** Architected a distributed video data processing pipeline using **Apache Spark** and **Argo Workflows**, cleaning **10,000+ hours** of raw video with frame-level and video-level filters to curate high-quality training datasets.
 - **AI Avatar & Gesture Synthesis:** Designed and deployed deep learning models for avatar realism, gesture synthesis, and speech-to-video synchronisation, resulting in a **30% increase in user engagement metrics**.
 - **Generative AI R&D:** Led research into state-of-the-art generative AI and diffusion-based architectures, continuously improving avatar fidelity and synthesis realism in production.
- **Intelligent Machines Lab, ITU** Lahore, Pakistan
Research Assistant Jun 2021 – Nov 2022
 - **3D Reconstruction & NeRF:** Developed PyTorch-based deep learning models for 3D reconstruction, NeRF preprocessing, and point cloud processing, improving reconstruction accuracy by **15%**.
 - **Camera Pose Estimation:** Designed and implemented camera pose estimation algorithms for 3D vision pipelines, achieving **90% accuracy** on benchmark datasets.
 - **Research Publication:** Co-authored a research paper on novel 3D point cloud upsampling methods, submitted to **CVPR** (Conference on Computer Vision and Pattern Recognition).

KEY PROJECTS

- **AI Avatar Fidelity & Realism Improvement (2025–2026):** Redesigned the core avatar generation pipeline architecture at Colossyan, improving **lip-sync accuracy** and overall avatar fidelity by **20%**, directly enhancing production video realism. Tech: Python, PyTorch.
- **Large-Scale Distributed Vision Data Processing Pipeline (2025):** Architected and deployed a distributed data processing pipeline at Colossyan using **Apache Spark** and **Argo Workflows** to clean **10,000+ hours** of raw video data using 6 frame-level and video-level filters, yielding a curated **20% high-quality subset** for downstream model training. Tech: Python, PyTorch, Spark, Argo Workflows.
- **AI-Powered Video Interview Proctoring System (2024–2025):** Built a real-time AI proctoring system integrating **deepfake detection**, gaze tracking, and facial behaviour analysis for automated interview integrity verification. Tech: Python, PyTorch, OpenCV.
- **Cancerous Blood Cell Detection (2024):** Developed a **YOLOv11**-based deep learning model for real-time detection of cancerous blood cells, achieving **95% accuracy** on microscopic image datasets. Tech: Python, PyTorch, YOLO.
- **Real-Time Video Background Matting (2023):** Optimised AI model for seamless real-time background removal, delivering measurable gains in visual quality and inference efficiency by **20%**. Tech: Python, PyTorch, OpenCV.
- **Neural Radiance Fields Without COLMAP (2022):** Pioneered a novel **NeRF-based 3D reconstruction** approach eliminating Structure-from-Motion (SfM) preprocessing, enabling reconstruction of texture-poor surfaces without COLMAP dependency. Tech: Python, PyTorch.

EDUCATION

- **Information Technology University of the Punjab** Lahore, Pakistan
Bachelor of Science – Computer Science *Sep 2018 – Jun 2022*
Relevant Modules: *Machine Learning, Computer Vision, Deep Learning, Data Structures & Algorithms, Parallel & Distributed Computing, Robotics, Artificial Intelligence*

ACHIEVEMENTS & AWARDS

- **2nd Place Nationally** – IEEE Xtreme Programming Competition, Institute of Electrical and Electronics Engineers (2021)
- **2nd Place** – Superior Technical Symposium (STS) Programming Competition, Superior University Lahore (2021)
- **Top 5 Finalist** – Ignite CyberSecurity Hackathon Pakistan, Speed Programming Track (2022)

LEADERSHIP & EXTRACURRICULARS

- **Event Management Lead – Google Developer Student Club ITU** Lahore, Pakistan
Organised technical events, developer workshops, and community talks. *2021*
- **Membership Team Lead – IEEE ITU Student Branch** Lahore, Pakistan
Led membership drives and coordinated IEEE chapter activities for student engineers. *2020*
- **Teaching Assistant – Parallel & Distributed Computing, ITU** Lahore, Pakistan
Mentored students, led problem-solving sessions, and supported course delivery. *2021 – 2022*

LANGUAGES

- Urdu – Native
- English – Professional Proficiency (C1) | TOEFL iBT: 103/120
- German – Basic (A2)

References available upon request.